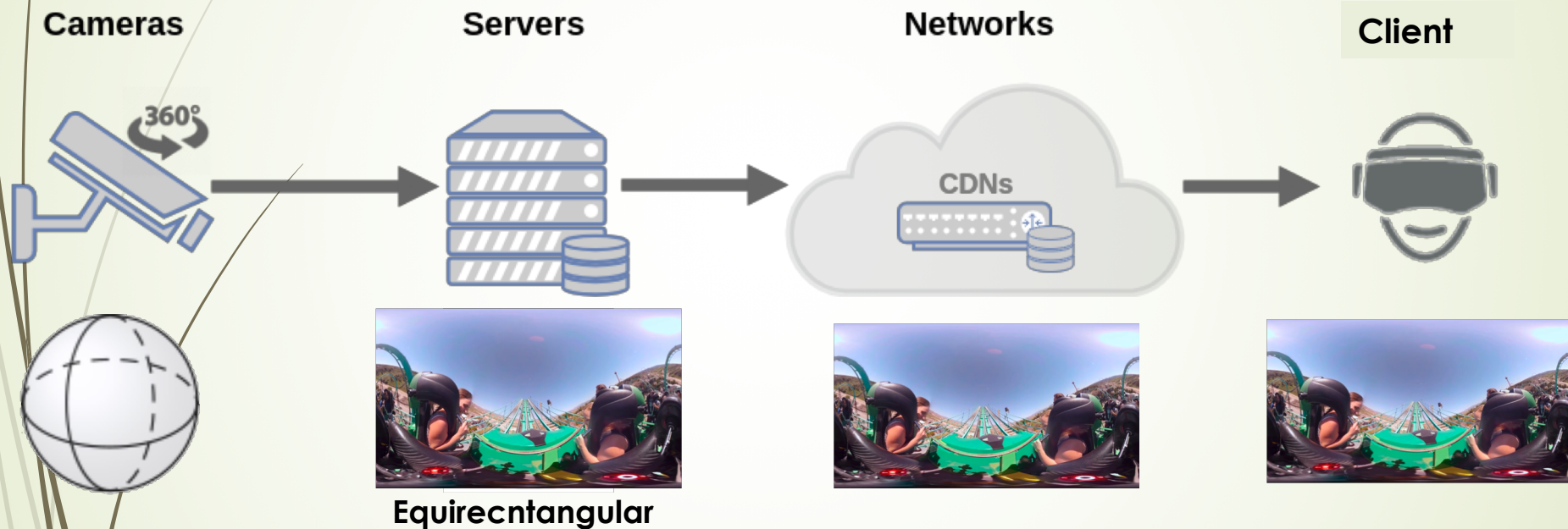


Smart Decisions Need to be Made by DASH Client



- Which quality level to request?
- Which tiles to request?
- When to request?

Programming Assignment #1

- You will modify the GPAC MP4Client to request tiled segments at different quality levels
- Please submit a PDF report
- If a demo is needed, I will let you know



Agenda

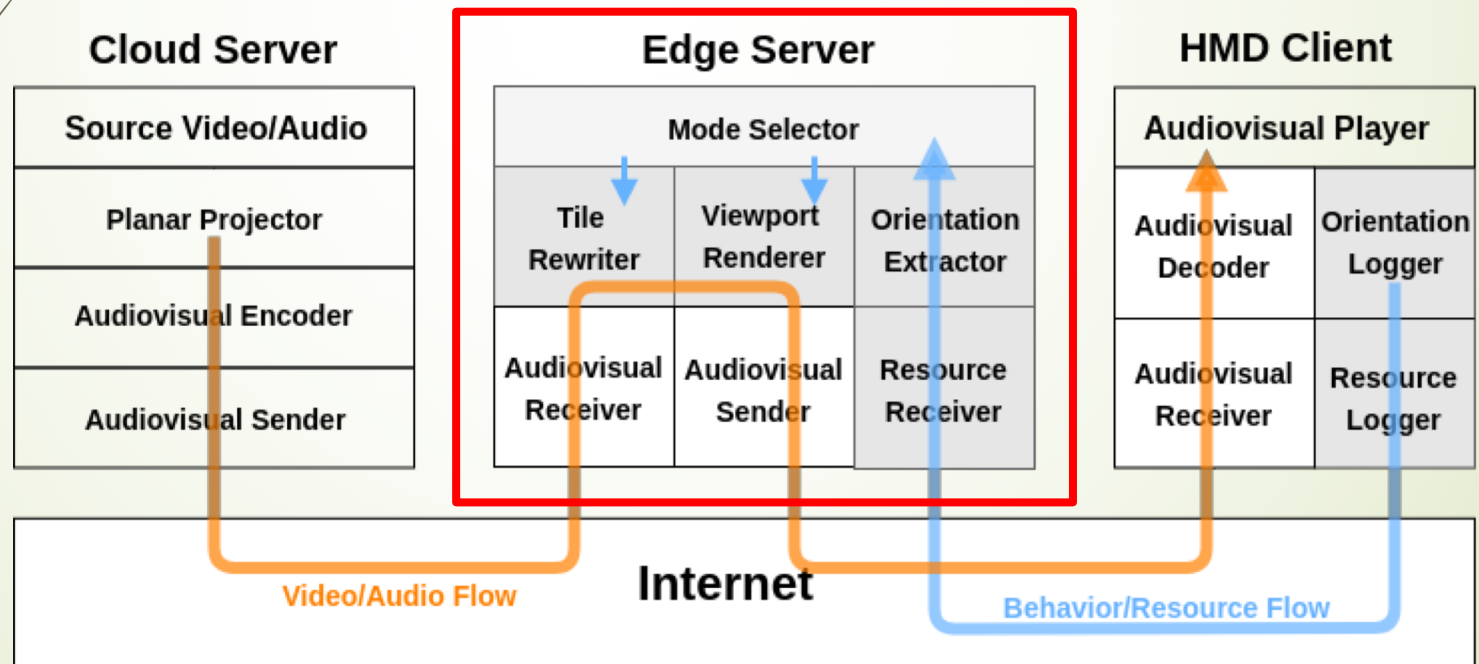
- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges

Why Edge Servers are Useful for Large-Scale 360 Video Streaming?

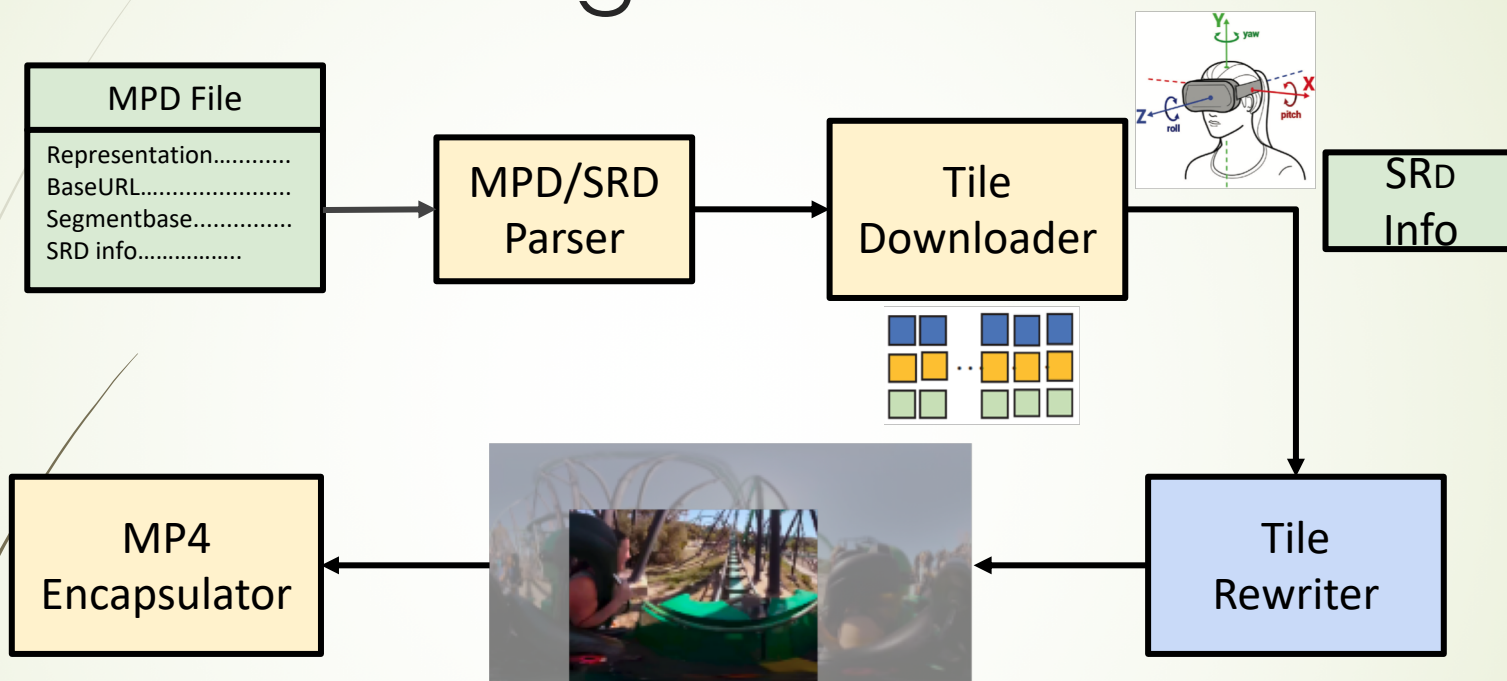
- ▶ High bandwidth consumption
 - ▶ Tiled streaming, but we want to cache some common tiled segmented at the **edge server**
- ▶ Latency sensitive
 - ▶ Tiled segments need to be near-by (at the **edge server**)
 - ▶ For weaker HMDs, computations can be done at the **edge server**
- ▶ Heterogeneous HMDs
 - ▶ How to allocate bandwidth (downstream from the **edge server**) to individual clients

Edge Servers Help to ...

- Tile Rewriting (TR): Merge tiles into a single video stream
- ViewPort Rendering (VPR): Generate two views for HMDs



Tile Rewriting: Merging Tiles at the Edge Server



- Parse Media Presentation Description (MPD) (with Spatial Representation Description (SRD) info)
- Download tiles at different qualities
- Combine tiles into a single HEVC bitstream based on SRD info
- Encapsulate the HEVC bitstream into an MP4 container